

# PulseDiscover

Interview Defense Dossier

Offline, honesty-gated recommender decision system — 2.56M Goodreads interactions

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**Status:** V1 [gold\\_candidate 8.9](#) (locked, terminal) · V2 lane [offline gold-complete](#) (RiskFrame 9.1) · serving API **deployed to Google Cloud Run**.

**Live endpoint:** <https://pulsediscover-serving-98058433335.us-central1.run.app> (/health → ready:true, /recommend → source:als).

**Reading rule:** every number here is offline and protocol-tagged. Served-c2 metrics and V1 d3applus metrics are **different protocols and not comparable** (see §6). No online-lift claim appears anywhere.

# 1 Project Summary

PulseDiscover is an offline, honesty-disciplined **recommender decision system** built on a real 2.56M-interaction Goodreads fantasy/paranormal dataset (94k users, 42k items). It does not chase a single metric — it **decides which retrieval, serving, fallback, fusion and exposure policy should ship, and proves which must not**, by catching offline-metric bias, candidate-coverage gaps, latency-quality trade-offs, cold-start degradation, and catalog concentration before they would reach users.

It was built as a gated experimentation program (gates G11–G31) under strict claim discipline: every claim maps to an artifact, every metric carries a protocol tag, and negatives are documented rather than buried. V1 established the offline modelling floor; the V2 lane (G22–G31) added production-shaped serving, cold-start and exposure governance, a learned ALS+semantic fusion ranker, executed off-policy evaluation, and a Search/IR retrieval front end — then the serving path was deployed to Cloud Run.

**What it is not:** not an online/production-quality system, not a solved-cold-start system, not a fairness-certified system, and not an LLM recommender. Those boundaries are explicit and load-bearing.

## 2 System Architecture

A two-stage retrieve→rank pipeline wrapped in a decision/serving spine. Every stage is versioned on the **served c2 ALS model**; V1's d3aplus tuning model is quarantined to V1 evidence only.

| Stage                   | What it does                                                                        | Artifact / evidence                    |
|-------------------------|-------------------------------------------------------------------------------------|----------------------------------------|
| 1. Candidate generation | ALS (warm) + semantic content (MiniLM) + popularity + co-occurrence                 | c2_als.pkl, g27 semantic index         |
| 2. Retrieval            | FAISS top-K by inner product; FlatIP exact (default), HNSW (scale), IVF rejected    | src/serving/faiss_retriever.py (G22)   |
| 3. Fusion / rank        | RRF heuristic fusion; LightGBM LambdaMART learned fusion over ALS+semantic features | src/ranking/g28_fusion_ranker.py (G28) |
| 4. Rerank               | Config-flagged head-cap exposure rerank (default OFF)                               | recommender_service.py (G26B)          |
| 5. Fallback             | unknown/sparse/error → popularity; never empty (0% empty verified)                  | fallback tree (G23/G24)                |
| 6. Serving              | FastAPI: /recommend /health /metadata /metrics; versioned loader; structured logs   | src/serving/api.py (G23)               |
| 7. Logging              | Per-request JSON; OPE-ready propensity schema designed                              | G26A schema, G29 logging               |
| 8. Evaluation           | Cohort recall, exposure (Gini/coverage), IPS/SNIPS/DR, NDCG/MRR                     | g24–g31 reports                        |
| 9. Decision             | ship / hold / reject per gate, behind a locked claim ladder                         | control tower + claim boundary         |

**Deployed:** the ALS+FlatIP serving path runs on Google Cloud Run (512 MiB / 1 vCPU / min-instances 1 / concurrency 8, exact mode, HNSW off). A lean image (only c2\_als.pkl + a 4 MB precomputed assets bundle) loads in ~0.8 s and serves live HTTPS.

### 3 Fundamentals Bridge — concept extraction

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Each technique, from first principles, with its role in PulseDiscover.

#### ALS (implicit-feedback matrix factorization)

Minimises  $\sum c_{ui}(p_{ui} - x_u \cdot y_i)^2 + \lambda \|\cdot\|^2$  with confidence  $c=1+\alpha r$ , solved by alternating closed-form ridge regressions. Captures collaborative co-read structure directly. **Role:** the warm-retrieval floor and the served model.

#### FAISS (ANN retrieval)

Sub-linear top-K over item vectors. FlatIP = exact exhaustive inner product; HNSW = navigable small-world graph (efSearch trades recall/latency); IVF = coarse quantiser (nprobe). **Role:** serving retrieval — FlatIP exact default, HNSW as a scale option, IVF rejected.

#### SASRec (self-attentive sequential rec)

Causal self-attention over the user's item-sequence to predict the next item; full-softmax vs sampled-softmax loss matters. **Role:** tested to convergence as the deep-model challenger — it lost to ALS (an honest negative).

#### LambdaMART (learning-to-rank)

Gradient-boosted trees optimising a listwise NDCG surrogate with per-query (per-user) groups. **Role:** the learned fusion ranker over ALS+semantic+popularity candidate features (G28); also the V1 within-candidate ranker.

#### IPS / SNIPS (off-policy estimators)

IPS =  $\text{mean}[1\{a=\pi_e(x)\} / \pi_{\log}(a|x) \cdot r]$  — unbiased but high-variance importance weighting. SNIPS self-normalises by  $\sum$  weights to cut variance at a little bias. **Role:** executed offline in G29 to estimate a policy's value from logged data; SNIPS is the recommended estimator.

#### RRF (reciprocal rank fusion)

$\text{score}(i) = \sum_s 1/(c + \text{rank}_s(i))$  across sources — score-scale-free, training-free rank combiner. **Role:** the best non-learned fusion of ALS and semantic candidates; chosen for robustness, not a recall win.

#### Position bias (and PAL)

Users click/consume top-ranked items partly because of position, not relevance; naive click-recall is therefore confounded. Position-aware learning (PAL) / IPS de-bias examination from relevance. **Role:** PulseDiscover does position-bias-aware *evaluation* (a two-lane OPE diagnostic) — not an in-ranker correction (that is documented future work).

## 4 Decision Log — trade-offs with rationale

| Decision                                                   | Rationale                                                                                                                                                                                                                                       | Boundary kept                                                 |
|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| <b>ALS over SASRec</b> as the warm floor                   | A canonical full-softmax SASRec was trained to convergence and still scored below ALS (R@20 0.065 vs 0.085). On short, sparse book histories the dominant signal is collaborative co-read, which MF captures directly; depth was not the lever. | "a converged deep model lost" — not "deep models are useless" |
| <b>FlatIP exact as default</b> (HNSW only as scale option) | FlatIP is lossless and ~47% faster than numpy brute force at this scale; HNSW ef64 is near-lossless (overlap 0.992, ~2.4x) but approximate. Correctness-by-default; opt into approximation explicitly.                                          | latency, not quality — FAISS never improves recommendations   |
| <b>RRF = robustness, not a recall win</b>                  | RRF recovers cold-start and ~4x coverage at ~95% of ALS relevance, but does not beat ALS-only on warm recall. Framed as robustness/coverage, never as a headline lift.                                                                          | no "hybrid beats both lanes" claim                            |
| <b>IVF rejected</b> despite speed                          | Low-nprobe IVF kept gold Recall@20 flat while candidate-overlap@K vs exact collapsed to 0.28-0.78 — it returned a largely different (broken) candidate set. Speed that destroys candidate fidelity is a bug.                                    | overlap@K is a first-class acceptance check                   |
| <b>BM25 = query-by-document</b>                            | PulseDiscover has no free-text queries; BM25's "query" is the seed item's own text (more-like-this). This is a real IR pattern, so the search framing is honest without claiming a free-text search engine.                                     | seed-item, not free-text query search                         |

## 5 Three Real Failures (caught and corrected)

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The interview value of the project is the failures I caught myself.

### SASRec false convergence — the NaN val-loss patience bug

**What happened:** training early-stopped at ~6 epochs — looked converged. The validation-loss curve was all-NaN (masked-attention NaN on short/padded sequences), and a NaN comparison silently satisfied the patience criterion while train loss and eval-R@20 were still rising. **Revealed:** a monitor can fire "done" for the wrong reason; "early stop triggered" is not evidence of convergence. **Did:** rejected the run as invalid, rebuilt the monitor on an eval-R@20 plateau, and only declared convergence (v3, ~95 epochs) once the metric we care about flattened. **Demonstrates:** I never accept a convenient stop; I check why a signal fired.

### IVF — the metric can lie

**What happened:** FAISS IVF at low nprobe looked great on gold Recall@20, but candidate-overlap@K vs exact had collapsed to 0.28–0.78 — it returned a mostly different candidate set that happened to still contain the single gold item. **Revealed:** a single-gold metric is blind to most of the slate changing, so a downstream metric can stay flat while the thing it depends on silently degrades. **Did:** rejected IVF despite its speed; kept FlatIP/HNSW and made overlap@K a first-class acceptance check. **Demonstrates:** I instrument the mechanism, not just the headline number.

### ID-dtype silent zeroing — the pipeline ran clean with wrong numbers

**What happened:** book\_id is a string in the ALS index, but eval/train ids loaded as int64; every membership test compared int vs str keys, so every lookup missed and recall floored to ~0 — while the pipeline ran clean and produced plausible-shaped (wrong) numbers. **Revealed:** the most dangerous bugs don't crash; silent type coercion quietly degrades a metric to zero while everything looks healthy. **Did:** caught it with an overlap sanity check (the genre map had zero catalog-overlapping keys — impossible if types matched), normalised book\_id to str repo-wide. **Demonstrates:** I sanity-check intermediate artifacts instead of trusting a final number.

## 6 Eval Results — with honest claim boundaries

**Protocol warning (load-bearing):** the V1 **d3aplust** ALS tuning result (R@20 **0.0846**) and the V2 **served-c2** results (full-catalog single-held-out-gold, R@20 ~**0.0375**) are **different model builds and different evaluation protocols — they are NOT comparable**. I keep them in a model/protocol registry precisely so they are never conflated. V2 numbers are used for *relative* cross-policy/cohort structure, not as a headline that overturns V1.

| Result                      | Number                                               | Protocol tag                                  |
|-----------------------------|------------------------------------------------------|-----------------------------------------------|
| V1 ALS warm floor           | R@20 0.0846                                          | V1 d3aplust tuning — REAL_OFFLINE             |
| Canonical SASRec, converged | R@20 0.065 (lost to ALS)                             | V1 — PLOT_RECONSTRUCTED                       |
| Served ALS (exact)          | R@20 ~0.0375                                         | V2 served-c2 — NOT_COMPARABLE to V1           |
| FAISS FlatIP                | lossless, ~47% lower p95                             | V2 latency — latency only                     |
| HNSW ef64                   | overlap 0.992, ~2.4x                                 | V2 latency                                    |
| Cold-start fallback cost    | coverage collapse ~59x; personalization 0.68→0.001   | V2 served-c2                                  |
| Exposure concentration      | all policies Gini>0.97; popularity 1.0               | V2 — concentration, not fairness              |
| Semantic cold-start reach   | 54.6% of unseen golds (ALS/pop = 0)                  | V2 — reachability, offline                    |
| Learned fusion vs ALS-only  | test R@20 0.036 vs 0.022; cold 0.032 vs 0            | V2 — held-out test, 81 positives, directional |
| OPE estimators              | DR 0.0089 vs true 0.0106; IPS over-est; SNIPS stable | V2 — offline, synthetic props, proxy reward   |
| Search/IR retrieval         | BM25 R@20 0.0133 > dense 0.0067; hybrid 0.0117       | V2 — seed-item, single-relevant NDCG/MRR      |

Every cell above is offline. Small-sample caveats (e.g. 81 test positives in the fusion result) are stated, not hidden; the learned-beats-ALS claim is restricted to the offline test split and is directionally consistent across four model families.

## 7 Truth Boundary

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### Claimable (artifact-backed):

- Built an offline two-stage recommender decision system on 2.56M real Goodreads interactions.
- ALS is the warm-retrieval floor — earned by beating a *converged* SASRec (documented honest negative).
- FAISS latency-quality frontier: FlatIP exact lossless ~47% faster; HNSW near-lossless; IVF rejected for overlap collapse.
- Production-like FastAPI+FAISS serving with versioned loader, fallback tree, logging, monitoring — and it is now **deployed to Cloud Run serving live HTTPS**.
- Measured cold-start fallback quality and catalog exposure concentration; added a semantic content lane that uniquely reaches cold-start items.
- A LightGBM LambdaMART fusion beat ALS-only on the held-out test split and recovered cold-start (offline, small positive set).
- Executed the OPE pipeline (logging + IPS/SNIPS/DR) and validated estimators recover a known value offline.

### Forbidden (never claimed):

- Online lift / engagement / A/B impact — no live experiment was run.
- Production *quality* or "served real users to effect" — deployed and callable is true; business impact is not.
- Solved cold-start, solved long-tail discovery, or fairness certified.
- LLM recommender / semantic taste understanding — the semantic lane is embedding-based candidate generation.
- Learned ranker beats ALS beyond the offline test split; real off-policy value (props were synthetic, reward a proxy).
- Any cross-protocol comparison of V1 d3aplay and served-c2 numbers.

## 8 Hard Q&A

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### Q. Why did SASRec lose to ALS?

A. Book next-item is weakly sequence-predictable; the dominant signal is collaborative co-read, which MF captures directly. I trained a canonical full-softmax SASRec to convergence (~95 epochs, eval-R@20 plateau) and it still scored below ALS — depth was not the lever on short, sparse histories.

### Q. Was the SASRec just undertrained?

A. No. I monitored convergence on eval-R@20 (after catching a NaN val-loss false-stop), and the full-softmax-vs-sampled-softmax gap (~6.7x) showed the loss objective was the real lever — and even fixed, it didn't beat ALS.

### Q. What's the OPE variance problem and how did SNIPS help?

A. Raw IPS uses importance weights  $\pi_e/\pi_{\log}$ ; at modest effective sample size a few large weights blow up the variance (my IPS over-estimated ~2x). SNIPS self-normalises by the sum of weights, trading a little bias for much lower variance — it's my recommended estimator when no reward model exists; DR is lowest-bias when a calibrated reward model does.

### Q. Why was DR negative the first time?

A. The DR reward model was a class-balanced logistic that calibrated toward ~0.5 vs the true ~0.01 base rate, so it massively over-predicted and broke the DR baseline. I caught it from the impossible negative value, switched to a base-rate-calibrated model, and DR then recovered the known value best of the three.

### Q. What is the real cost of cold-start here?

A. Not a recall drop — recall is actually non-monotonic (cold users' next item is more popularity-predictable, so it's easier to retrieve). The real cost is that popularity fallback collapses catalog coverage ~59x and personalization from 0.68 to ~0.001. I reframed the metric around what actually degrades.

### Q. Why did you reject IVF if it was faster?

A. At low nprobe IVF kept gold Recall@20 flat but candidate-overlap@K vs exact collapsed to 0.28–0.78 — it served a largely different, broken candidate set. Speed that silently destroys candidate fidelity is a recall bug, so I kept FlatIP/HNSW and made overlap@K an acceptance check.

### Q. Why did BM25 beat dense on this data?

A. On short text (title + genre shelves), exact lexical overlap — shared series names, shared genre words — carries more signal than a 384-d embedding that compresses and blurs exactly those rare disambiguating tokens. Embeddings win on long, paraphrase-heavy text; lexical wins on short, keyword-dense text. So I reported BM25 as the strongest single lane and hybrid as robustness, not a win.

### Q. So is your hybrid retrieval better?

A. On recall, no — RRF hybrid sits below BM25 alone on this seed-item task. Its value is robustness: it never collapses to either lane's weakness and it reaches cold-start. I don't claim hybrid beats both lanes.

### Q. Did the learned ranker actually beat ALS?

A. On the held-out test split, yes (R@20 0.036 vs 0.022) and it recovered cold-start (0.032 vs 0), consistent across four model families — but on only 81 positives with a single held-out gold, offline. I treat it as directional, not a production claim, pending OPE/online evidence.

### Q. Why does fusion help at all?

A. ALS and semantic candidate sets are almost disjoint — they overlap on only 6,749 of ~880k candidates. A learned ranker can pick the best items across both complementary sources, which is what lifted test recall and cold-start.

**Q. What does the live Cloud Run endpoint actually serve?**

A. The served-c2 ALS model with exact FlatIP FAISS retrieval: GET /recommend returns top-K items with source tags and a graceful fallback (unknown user → popularity, never empty). /health reports loader readiness. It is the real serving path — not an online-lift experiment. "Deployed and serving HTTPS" is true; "improved engagement" is not.

**Q. Is any of this production / online?**

A. No. Everything is offline by scope. The serving API is deployed and callable, but there are no real users, no logged real-traffic propensities, and no A/B. That boundary is documented as the explicit next step, not hidden.

**Q. What would falsify your headline claims?**

A. ALS-floor: a properly-tuned sequence/graph model beating it on this exact protocol. Fusion-win: it not holding on a larger positive set or a different relevance target. Semantic value: cold-start reachability not translating to any online engagement. OPE: ESS too low / no overlap. I state these kill conditions rather than hide them.

## 9 What I'd Build Next

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With production access, in priority order:

- **Real-traffic propensity logging** — populate the OPE schema (already designed) with live  $\pi_{\log}$ , position, exploration bucket and reward, to turn IPS/SNIPS/DR from a validated methodology into real off-policy value estimates.
- **Canary + live A/B** of the fusion policy and the head-cap exposure rerank, with guardrail metrics (relevance, coverage, Gini, fallback rate, p95) wired to alerts — the only way to claim online lift.
- **In-ranker position-bias correction (PAL / IPS-weighted LTR)** — upgrade position bias from *evaluated* to *corrected*.
- **Productionise the lexical lane** (Lucene/Elasticsearch/bm25s) so BM25 is sub-millisecond, and persist the FAISS index to cut cold-start build time.
- **Two-tower / GNN retrieval and graded-relevance labels** — only once there's signal/scale to justify them; today they're deferred, not silently skipped.

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Companion material: docs/PULSEDISCOVERY\_INTERVIEW\_KIT.md (pitch & claim ladder), docs/PULSEDISCOVERY\_UNIFIED\_DEFENSE\_KERNEL.md (full method cards), docs/PULSEDISCOVERY\_FAILURES\_AND\_HARDENING.md (failures + serving stress test), docs/G26A\_MODEL\_PROTOCOL\_REGISTRY.md (V1/V2 metric registry).